

STAAD.PRO COURSE CONTENT

Duration- 4-6 Weeks

WEEK - 01

Introduction & Basic Modeling

- Introduction to Structural Analysis
- Overview of STAAD.Pro environment & interface
- Coordinate systems and units
- Creating nodes, beams, and elements
- Understanding geometry creation methods
- Practice: Create a simple frame model

WEEK - 02

Loads & Load Combinations

- Assigning Dead, Live, Wind, and Seismic loads
- Load cases and load combinations (IS 875 & IS 1893)
- Load generation: Floor load, wind load, seismic load
- Practice: Apply load combinations on a G+1 residential building

WEEK - 03

Analysis & Design of RCC Structures

- Structural analysis: Static and dynamic
- Design of RCC elements: beams, columns, slabs
- Reinforcement detailing and design parameters
- Using IS 456:2000 for RCC design
- Practice: RCC frame building analysis and design

WEEK - 04

Steel Structure Design

- Introduction to steel design as per IS 800
- Truss modeling, loading, and analysis
- Design of steel members and connections
- Buckling and slenderness checks
- Practice: Industrial shed design with steel trusses

WEEK - 05

Foundation Design & Report Generation

- Isolated and combined footing design
- Introduction to STAAD Foundation Advanced
- Interpreting analysis results (deflections, forces, reactions)
- Generating calculation reports and drawings
- Practice: Design and report for a complete structure

WEEK - 06

Project Work & Advanced Features

- Capstone Project: RCC/Steel structure (G+3 building or industrial shed)
- Introduction to advanced topics: dynamic analysis, response spectrum
- Interoperability with AutoCAD & Revit
- Tips for optimizing models and reducing errors
- Final presentation & evaluation

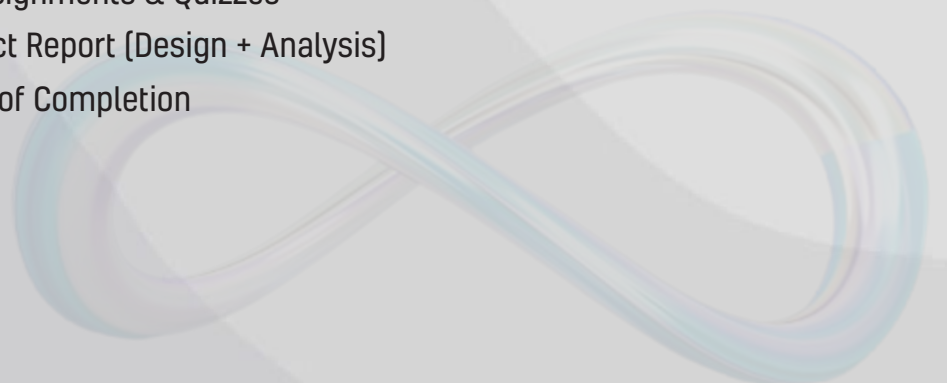
Tools & Standards Covered

- **Software:** STAAD.Pro Connect Edition, STAAD Foundation
- **Codes:** IS 456, IS 800, IS 875, IS 1893, AISC, Eurocode (optional)
- **Integrations:** AutoCAD (for import/export), Excel (for load calculation)

WEEK - 08

Course Deliverables

- STAAD Model Files & Load Templates
- Lecture Notes and Video References
- Weekly Assignments & Quizzes
- Final Project Report (Design + Analysis)
- Certificate of Completion



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